

PROJECT	DOCUMENT	REVISION	DATE
APOLLO TRACK POD	FABRICATION BLUEPRINT	001A	2026-07-10
UNITS	SHEETS		
MM [IN]	7 + SCHEDULES		

MODIFICATION TYPE – ARTICULATED SPLIT-FRAME

ARTICULATED "SPLIT-FRAME" SUSPENSION MOD

Converts a rigid bolt-on rubber-track pod (spoked drive sprocket on a hex axle, two bearing-mounted lower idlers, lugged rubber band) into an independently articulating suspension: a central pivot below the drive hub, a leading and a trailing arm, and two internal coil-over shocks. Every fabrication dimension is derived from eight measurements you take off your actual pod — Sheet 0 first, then cut metal.

§0 READ THIS BEFORE ANYTHING ELSE

The pod you bought (photos: sprocket + two idlers + rubber band, sold as a wheel-replacement track for mowers / small ATVs / mobility platforms) has **no external steel frame**. The idlers hang off an internal molded/plate structure hidden behind the sprocket, and dimensions vary between sellers even when the listings look identical. That structure is exactly what this mod removes and replaces, so:

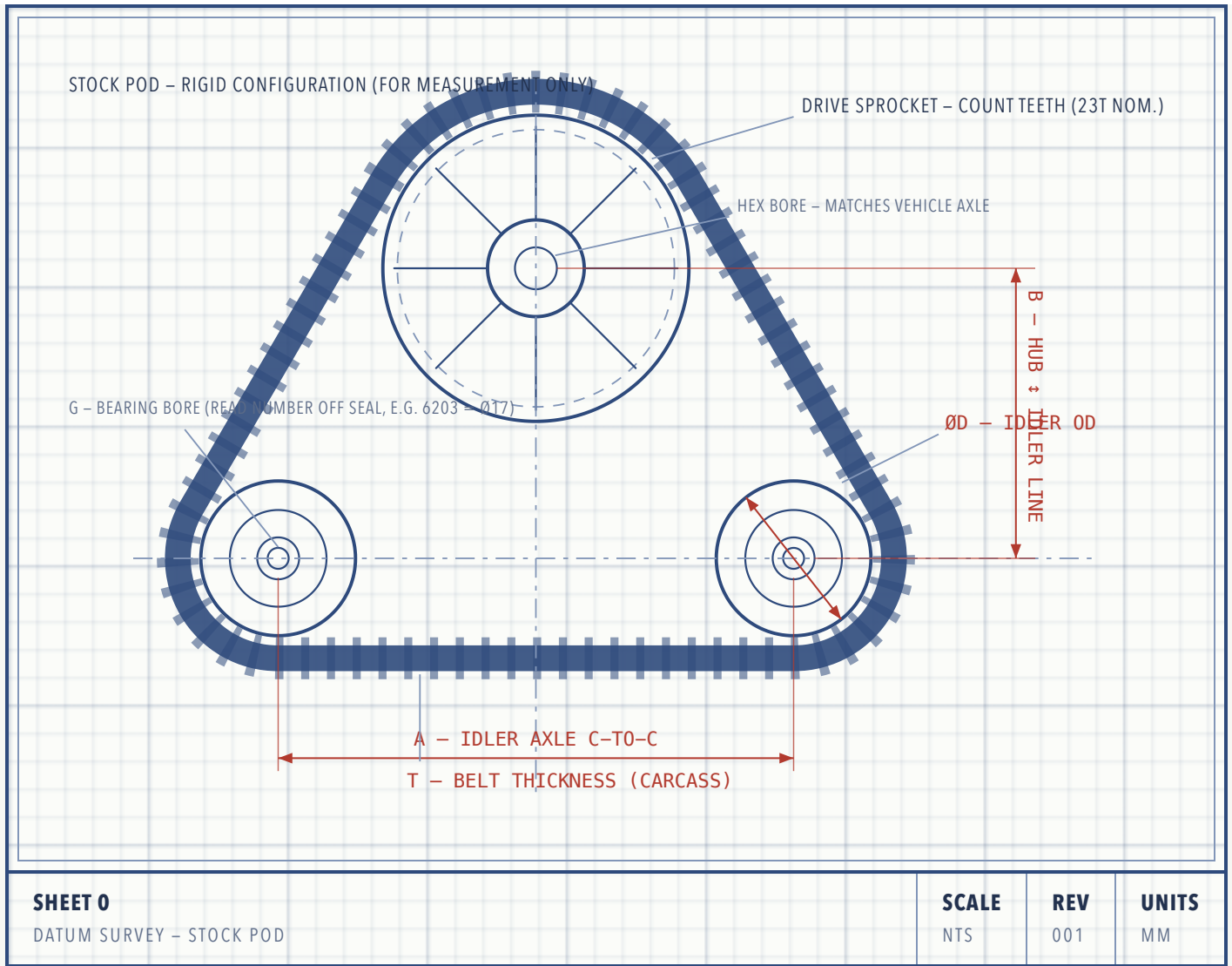
- **This drawing set is parametric.** Datums A–J on Sheet 0 are measured off *your* pod with calipers and a ruler. Every dimension on Sheets 1–6 is a formula of those datums. Fill the table in, then transfer numbers to the drawings.
- **Disassemble one pod completely first.** Photograph the internal frame, the idler axle hardware, and — critically — however the pod stops itself from spinning around the drive axle (anti-rotation). Your carrier plates must reproduce that interface.
- **Build one pod, test it, then replicate.** Don't cut four sets of parts before the first pod has survived the Sheet-7 test protocol.

Track retention is the failure mode of this mod. A rigid triangular pod keeps the belt at constant wrap length. Once the arms articulate, the belt path length changes a few millimetres through travel and the belt *will* try to walk off the idlers under side load. The design compensates three ways, and all three are mandatory: travel limited to $\pm 15^\circ$, spring preload keeping the belt tensioned through droop, and the trailing-arm tensioner (Sheet 6) set correctly. Deep-lug tracks like yours help; they are not sufficient alone.

Anti-rotation is not optional. With the axle spinning the sprocket, the entire pod body experiences drive torque. The carrier plate assembly must be restrained to the vehicle chassis by a torque link (Sheet 4, item ⑧), exactly as the stock pod is. Omit it and the whole pod rotates on the first throttle input.

SHEET 0 DATUM MEASUREMENTS – TAKE THESE OFF YOUR POD

Set the pod on a flat floor, belt tensioned as delivered. Measure with calipers where possible. Record everything in the table; the formula column on later sheets consumes these letters.



DATUM TABLE – FILL IN WITH CALIPERS

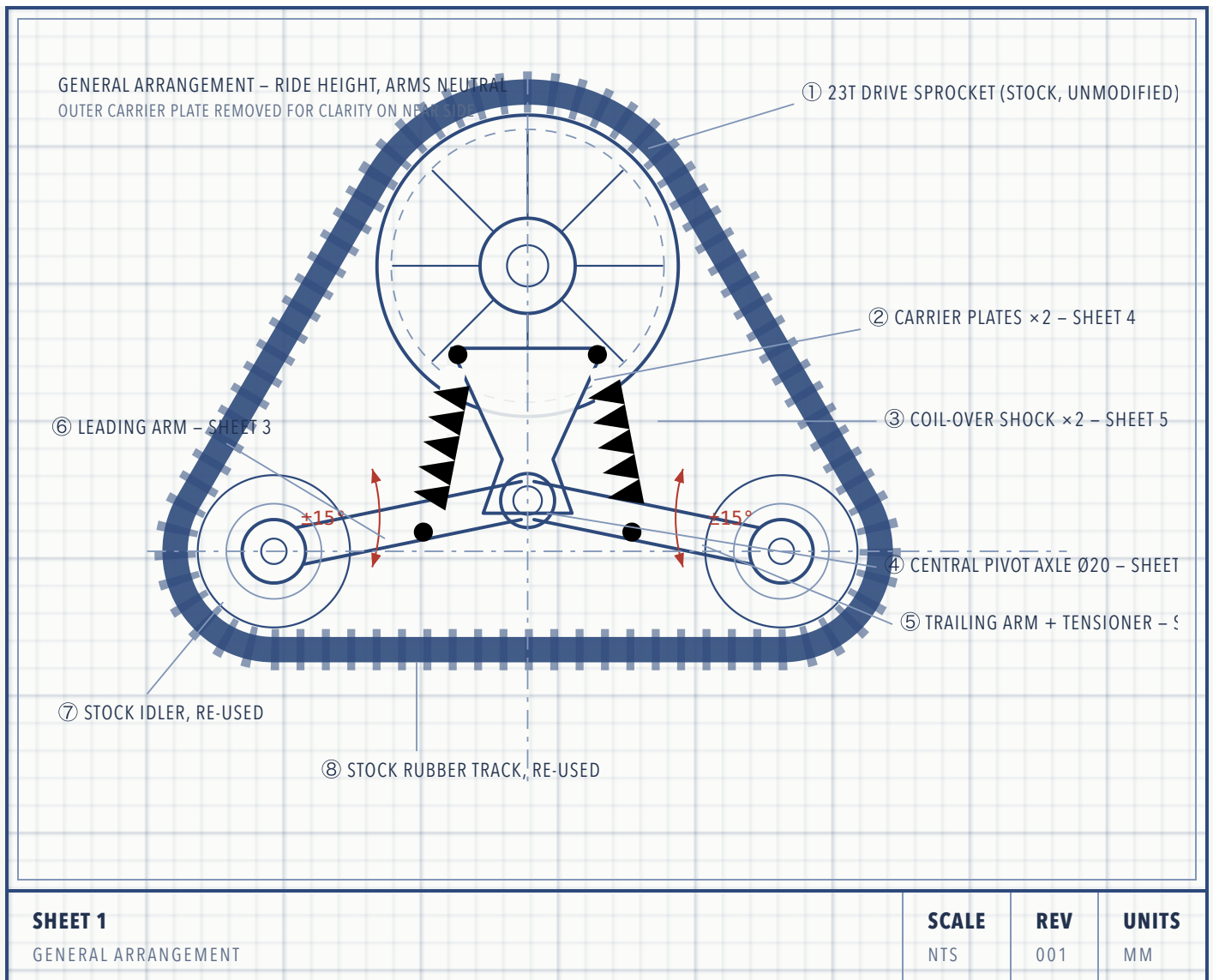
DATUM	WHAT TO MEASURE	HOW	TYPICAL*	YOURS
A	Idler axle centre-to-centre distance	Bearing centre to bearing centre, pod on floor	230–300	_____
B	Drive-hub centre height above idler axle line	Square + ruler from floor; B = hub height – idler height	150–200	_____
D	Idler wheel outside diameter	Calipers across the wheel	90–120	_____
G	Idler bearing bore (axle Ø)	Read bearing number off seal: 6202→Ø15, 6203→Ø17, 6204→Ø20	15–20	_____

DATUM	WHAT TO MEASURE	HOW	TYPICAL*	YOURS
F	Belt inner width between guide lugs	Calipers inside the two lug rows	60–90	_____
H	Idler hub width (bearing-to-bearing over the wheel)	Calipers over the wheel hub	40–60	_____
T	Belt carcass thickness (not incl. lugs)	Calipers on a straight span edge	8–14	_____
J	Vehicle axle Ø / stock pod mounting interface	Disassemble one pod; photograph + measure everything between sprocket and idlers, incl. anti-rotation feature	–	_____

*Typical ranges for this class of pod — for sanity-checking only, never for cutting. Derived values used on later sheets: $P = B - 20$ (pivot drop below hub centre) and $C = \sqrt{(A/2)^2 + 20^2}$ (pivot-to-wheel arm length). The 20 mm term puts the pivot just above the idler axle line so the arms sit near-horizontal at ride height.

SHEET 1 GENERAL ARRANGEMENT – HOW THE MECHANISM WORKS

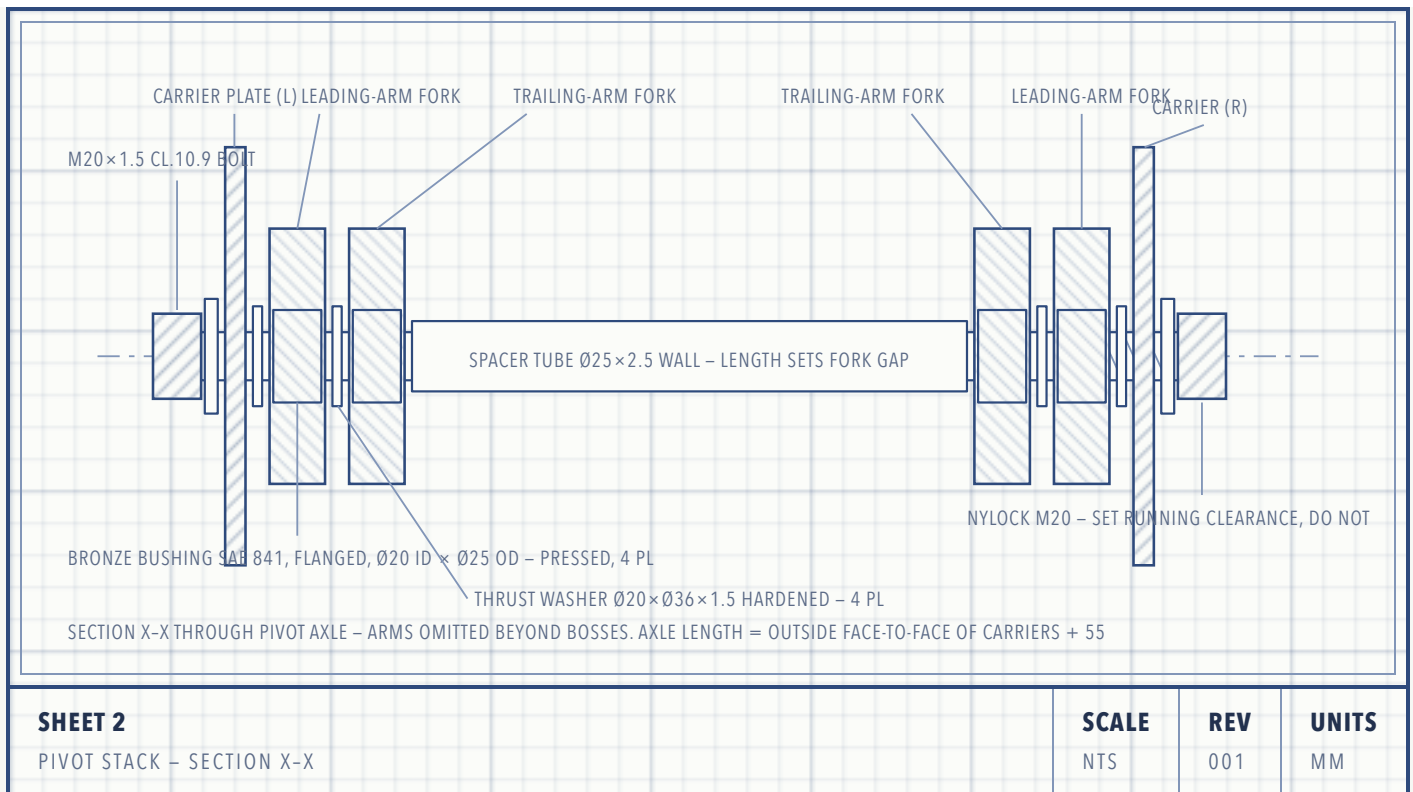
The rigid internal frame is replaced by two mirrored steel **carrier plates** that ride on the vehicle axle via their own bearings (they must not spin with it). A **Ø20 pivot axle** spans the plates directly below the hub. The **leading arm** (front idler) and **trailing arm** (rear idler) are fork-shaped weldments that swing on that axle like scissor halves. Each arm carries a **coil-over shock** reacting against tabs on the carrier plates. Wheel hits a bump → arm swings up → shock compresses. Each side is fully independent.



Arms drawn at neutral. Vertical wheel travel at $\pm 15^\circ$ is $C \cdot \sin 15^\circ \approx 0.26 \cdot C$ — for $C = 150$ mm that's $\approx \pm 39$ mm at each wheel, a huge improvement over the rigid pod's zero.

SHEET 2 CENTRAL PIVOT AXLE – STACK-UP SECTION

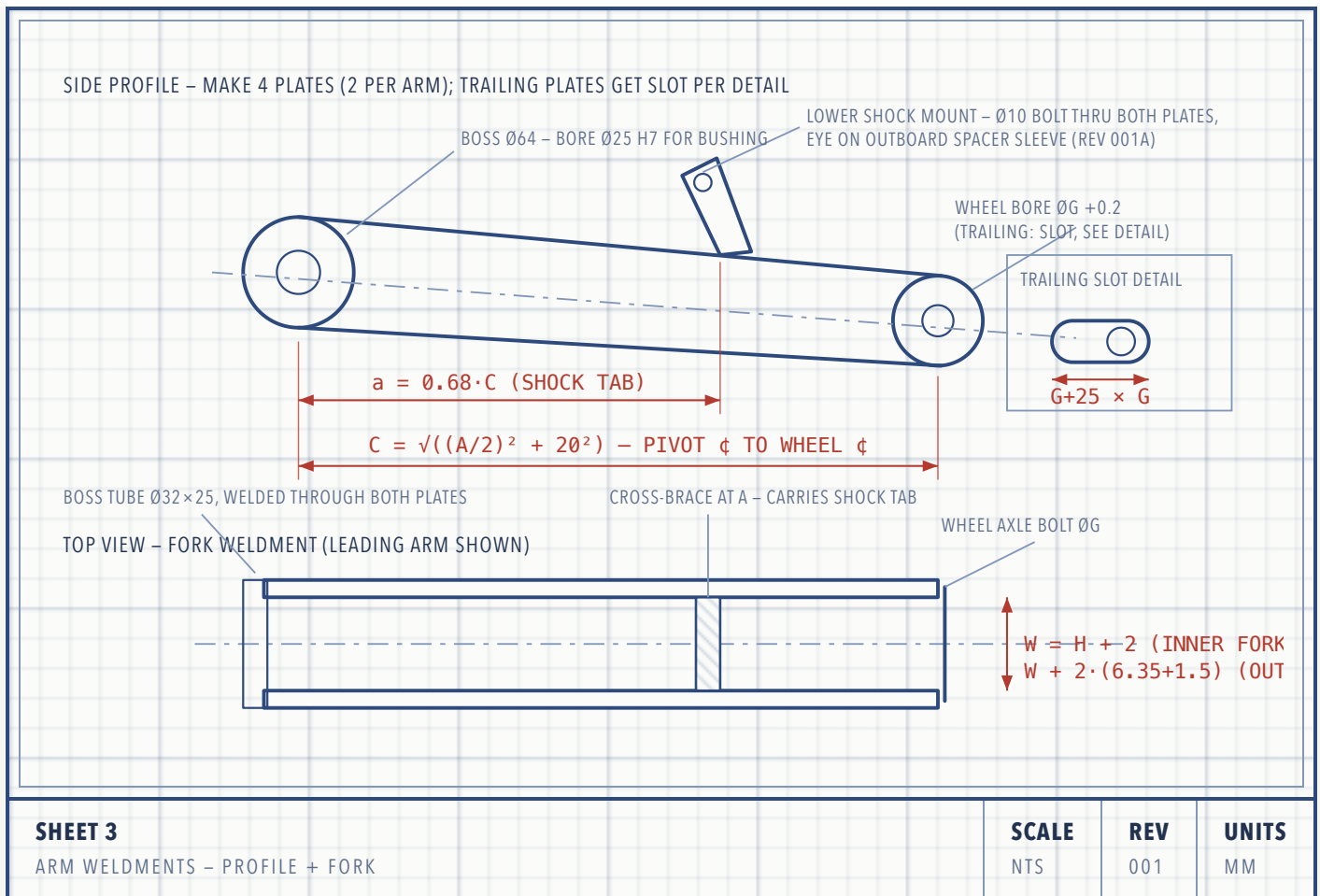
Both arms share one axle. To keep each idler wheel centred on the belt, each arm is a **fork** (two plates): the trailing arm is the narrow inner fork, the leading arm the wider outer fork, nested like scissors. Four bronze bushings — one pressed into each fork plate — carry the rotation. The axle itself is a partially-threaded M20 class-10.9 bolt clamping nothing: the nylock nut is set to leave running clearance, then the spacer tube takes the clamp load.



- Bushing seats in the fork plates: bore $\varnothing 25$ H7. Press bushings with a vise and a socket, never a hammer. Ream to $\varnothing 20$ sliding fit after pressing if they close up.
- Grease: lithium EP2 at assembly; add a cross-drilled grease passage + M6 zerk in each fork boss if you want serviceability (recommended — this joint lives in mud).
- The spacer tube length sets the lateral position of both forks: cut it so each idler wheel sits dead-centre in the belt (check against datum F before welding anything).

SHEET 3 ARTICULATING ARMS – LEADING & TRAILING

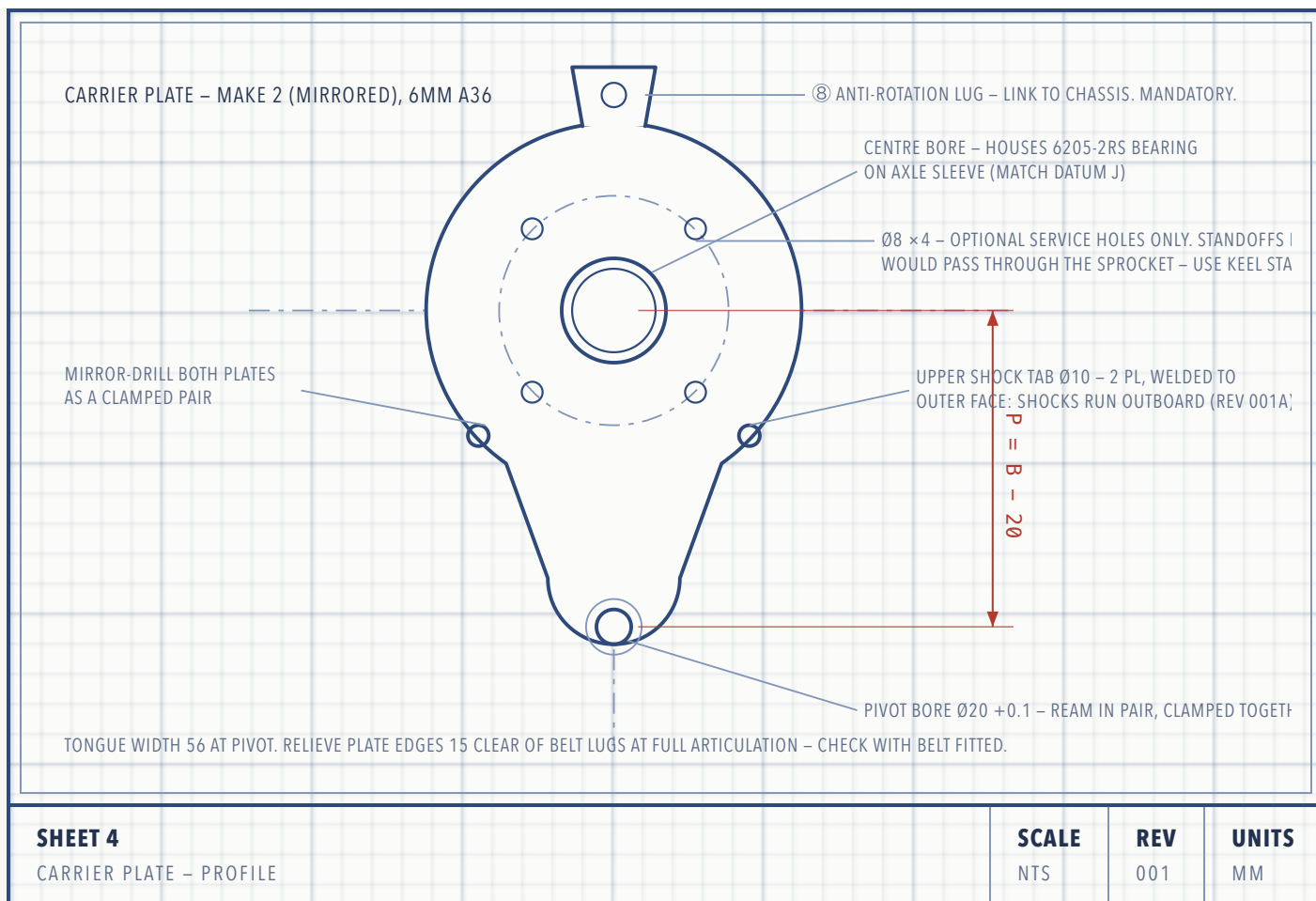
Each arm is a weldment of two profile plates (6.35 mm / ¼" A36 steel), a bushing boss tube at the pivot end, a cross-brace, and a shock tab on the brace. The two arms are identical except: the trailing arm's wheel bore is a **25 mm slot** for the belt tensioner (Sheet 6), and its fork is the narrower, inner one.



- **Arm length C is sacred.** At neutral (arms level) the wheel centres must land exactly where the stock frame held them, or the belt won't tension. Cut plates 1 mm long and file the wheel bores to final position with the belt fitted.
- Plate profile: 64 mm wide at the boss tapering to 52 mm at the wheel end is plenty for a <150 kg vehicle corner. Don't drill lightening holes on rev 001.
- Steel over aluminium: 6061-T6 loses ~40% strength in the weld HAZ and these arms are all welds. Use A36/S235 steel unless you're bolting, not welding.

SHEET 4 CARRIER PLATES – THE NEW BACKBONE

Two mirrored 6 mm steel plates replace the stock internal frame. Each has: a centre bore carrying a sealed bearing that rides on the vehicle axle (or on the stock pod's axle sleeve — copy datum J), the Ø20 pivot bore on the tongue below, upper shock-tab holes both sides, and an anti-rotation lug at the top that links to the vehicle chassis exactly as the stock pod did.



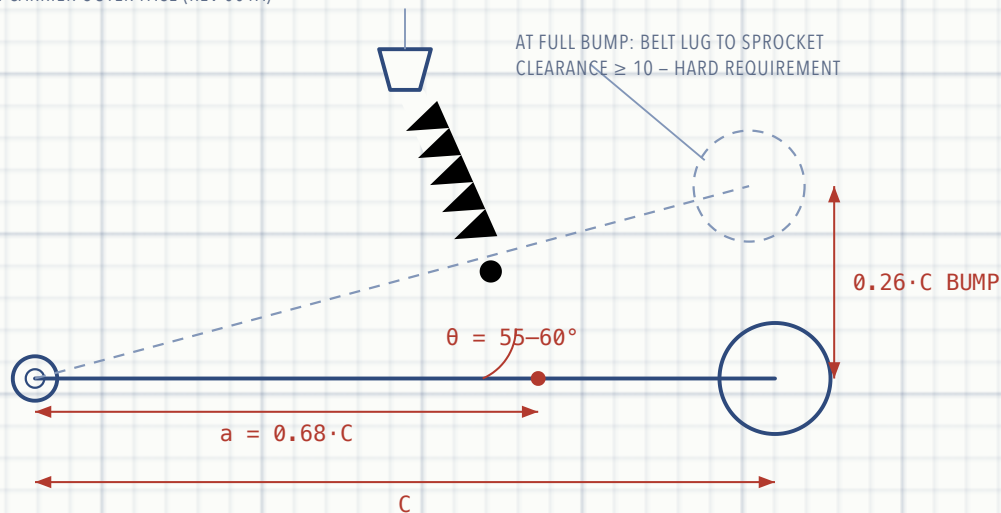
FIT-UP NOTE

The two carrier plates are joined into one rigid box by: the centre axle sleeve, the pivot axle assembly, and one M8 **keel standoff below the pivot** (rev 001a — standoffs at the bolt circle would pass through the sprocket's swept disc; the only clear tie points are on the centreline above the arms' swing or below it). Extend the tongue ~45 mm below the pivot to carry the keel. Weld nothing between the plates until the sprocket has been test-fitted spinning freely between them with ≥ 3 mm side clearance each face.

SHEET 5 SHOCK GEOMETRY, TRAVEL & SPRING SIZING

The shock mounts at $a = 0.68 \cdot C$ along the arm, inclined 55–60° to the arm. That placement is deliberate — it sets the motion ratio high enough that off-the-shelf 150 mm e-scooter/pit-bike coil-overs work without absurd spring rates.

UPPER EYE – TAB ON CARRIER OUTER FACE (REV 001A)



MOTION RATIO $MR = (A/C) \cdot \sin \theta \approx 0.68 \times 0.83 \approx 0.58$ • SHOCK STROKE USED $\approx 0.26 \cdot C \times 0.58$ (KEEP $\leq 80\%$ OF RATED STROKE)

SHEET 5

SHOCK GEOMETRY + TRAVEL

SCALE

NTS

REV

001

UNITS

MM / N·MM⁻¹

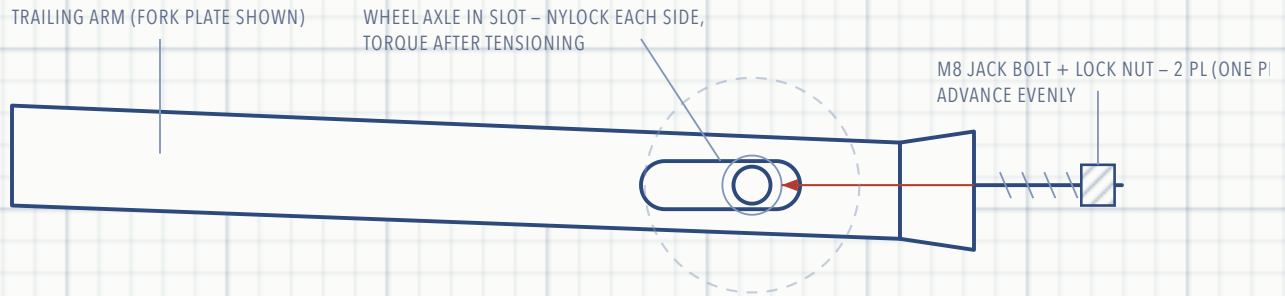
SPRING RATE – WORKED METHOD

1. Weigh the corner: put the vehicle's wheel-position on a bathroom scale, loaded (rider + cargo). Call it M kg $\rightarrow$ force $F = 9.81 \cdot M$ N. That corner load splits across the pod's two arms roughly 50/50 $\rightarrow F_{\text{arm}} = F/2$.
2. Wheel rate target: static sag should be 25–30% of total wheel travel. With $C = 150 \rightarrow$ travel ≈ 39 mm, sag ≈ 11 mm $\rightarrow k_{\text{wheel}} = F_{\text{arm}} / 11$ N/mm.
3. Spring rate: $k_{\text{spring}} = k_{\text{wheel}} / MR^2$ with $MR \approx 0.58 \rightarrow k_{\text{spring}} \approx 3.0 \times k_{\text{wheel}}$.

Example — 160 kg loaded vehicle on 4 pods $\rightarrow M = 40$ kg, $F = 392$ N, $F_{\text{arm}} = 196$ N $\rightarrow k_{\text{wheel}} \approx 18$ N/mm $\rightarrow k_{\text{spring}} \approx 54$ N/mm ≈ 310 lb/in. That's a normal heavy e-scooter / mini-moto spring. Buy adjustable-preload coil-overs, 150 mm eye-to-eye, ~ 35 mm stroke, and buy one spare spring one step stiffer and one softer.

SHEET 6 BELT TENSIONER – TRAILING ARM

The articulating geometry cannot hold belt tension by luck; the trailing wheel axle rides in the Sheet-3 slot and is pushed rearward by a jack bolt against a welded lug. This gives ~ 25 mm of tension range — enough to fit the belt slack and take up wear.



TENSION SPEC: 10-15MM BELT SAG AT MID-SPAN UNDER 5KG PUSH, ARMS AT NEUTRAL.
RE-CHECK AFTER FIRST HOUR OF RUNNING. A NEW BELT SEATS AND LOOSENS.

SHEET 6

TENSIONER - TRAILING AXLE

SCALE

NTS

REV

001

UNITS

MM

§7 BILL OF MATERIALS

#	ITEM	QTY/POD	SPECIFICATION
1	Pivot axle	1	M20×1.5 cl.10.9 bolt, part-threaded
2	Arm profile plates	4	6.35mm ($\frac{1}{4}$ ") A36/S235 steel
3	Carrier plates	2	6mm A36/S235, mirrored pair
4	Boss tubes	2	Ø32 OD × Ø25 ID DOM tube, 25 long

#	ITEM	QTY/POD	SPECIFICATION
5	Bronze bushings	4	SAE 841 flanged, Ø20 ID × Ø25 OD × 20
6	Thrust washers	4	Ø20 × Ø36 × 1.5, hardened
7	Spacer tube	1	Ø25 OD × 2.5 wall, cut to fit
8	Coil-over shocks	2	150mm eye-to-eye, ~35mm stroke, adj. preload
9	Shock mounts	2+2	2 upper tabs 6mm w/ Ø10 hole; 2 lower spacer sleeves Ø15×Ø10
10	Carrier bearings	2	6205-2RS (Ø25×Ø52×15) – adjust to datum J
11	Keel standoff	1	Ø16 tube + M8 through-bolt cl.10.9
12	Jack bolts	2	M8×40 + lock nuts

#	ITEM	QTY/POD	SPECIFICATION
13	Anti-rotation link	1	Copy stock pod's provision (datum J)
14	Hardware kit	—	Cl.10.9 (Gr8) bolts, nylock nuts, washers
15	Grease fittings	2	M6 zerk + EP2 lithium grease
—	Re-used from stock pod	—	Sprocket, 2 idlers + bearings, belt, axle hardware

§ 8 FABRICATION SPECIFICATIONS

CUTTING	Laser/waterjet preferred (send Sheet 3/4 profiles as DXF once dimensioned). Hand fabrication acceptable: plasma or 1 mm cutoff wheel, finish edges to bright metal, drill press only for bores — no hand drilling of bearing/bushing seats.
CRITICAL BORES	Bushing seats Ø25 H7 (step-drill then ream). Pivot bores in carrier plates reamed as a clamped pair so the axle is square. Wheel bores ØG +0.2.
WELDING	MIG, ER70S-6, 5 mm fillets. Boss tubes welded both faces. Shock tabs welded both sides — a single-side tab weld is the classic failure. No preheat needed on A36 at these thicknesses. Weld with bushings <i>out</i> (heat kills them); press bushings after paint.
SQUARENESS	Fork plates must be parallel within 0.5 mm over their length, or the idler runs angled and throws the belt. Tack, measure, weld in a staggered sequence, re-measure.

FINISH	Degrease, zinc-rich primer, topcoat. Mask every bore and bearing seat.
TORQUE	M8 cl.10.9 — 30 N·m · M10 — 60 N·m · M12 — 105 N·m · M20 pivot nylock — snug to zero end-float then back ⅛ turn (running fit, not clamped). Mark every fastener with paint-pen witness lines.

§9 ASSEMBLY SEQUENCE

- 1. STRIP THE DONOR POD.** Remove belt, idlers, sprocket. Photograph and measure the internal frame and anti-rotation feature (datum J) before discarding it.
- 2. PRESS BUSHINGS.** Four flanged bronze bushings into the fork-plate bosses (vise + socket). Verify the Ø20 axle turns by hand with no rock.
- 3. DRY-STACK THE PIVOT.** Thread onto the axle: carrier L → washer → leading fork → washer → trailing fork → spacer tube → trailing fork → washer → leading fork → washer → carrier R (Sheet 2 order). Cut the spacer so each idler centreline lands mid-belt (datum F). No welding yet.
- 4. BOX THE CARRIERS.** Fit the keel standoff and the centre axle sleeve + bearings. Spin the sprocket between the plates — ≥3 mm clearance each face, zero rub.
- 5. LOCATE SHOCK MOUNTS.** With arms level and shocks held in place at ~30% compression, mark positions: upper tabs on the carrier *outer* faces, lower eyes on Ø10 through-bolts with outboard spacer sleeves at $a = 0.68 \cdot C$, $\theta = 55\text{--}60^\circ$ (shocks run outboard of the carriers — rev 001a). Tack the tabs, articulate $\pm 15^\circ$ by hand checking nothing binds, then final-weld with bushings removed.
- 6. PAINT, THEN FINAL-ASSEMBLE** the pivot stack with grease, set the M20 nylock for free running, fit shocks.
- 7. FIT IDLERS** to arm forks — leading in the round bore, trailing in the slot, jack bolts backed off fully.
- 8. WRAP THE BELT.** Sprocket first, engage lugs, lever over the idlers with the trailing axle at its most-forward slot position (rubber levers only — no screwdrivers into the carcass).
- 9. TENSION.** Advance both jack bolts evenly to 10–15 mm mid-span sag, square the axle in the slot (equal bolt counts), torque the axle nylocks.
- 10. MOUNT TO VEHICLE.** Pod onto the hex axle, axle nut torqued, **anti-rotation link connected**, then run the Sheet-10 test protocol before carrying a rider.

§10 TEST PROTOCOL & SAFETY

1. **Bench articulation:** pod off the ground, cycle each arm full travel 50× by hand. No binding, no belt-lug contact with the sprocket at full bump (≥ 10 mm, Sheet 5), no shock coil-bind.
2. **Bench spin:** drive the pod unloaded at walking-pace RPM for 5 minutes. Watch belt tracking — it must not walk toward either edge. If it walks, the fork plates aren't parallel or the tension axle isn't square.
3. **Loaded creep:** vehicle weight on, no rider, walk-beside at < 5 km/h over a 50 mm plank. Confirm articulation and belt retention.
4. **Ridden test:** flat ground, low speed, then progressively rougher terrain. Stop and inspect after 10 minutes: retorque everything, re-check belt tension (new belts loosen), look for cracked tack welds and shiny rub marks.
5. **Retorque schedule:** after 1 h, 5 h, then every 20 h. Witness-mark lines make this a 2-minute glance.

SAFETY

Tracked pods are finger-traps: never work near a powered, moving belt; the sprocket-to-belt nip point will pull a glove in instantly. Wear eye protection when levering the belt on — it stores real energy. And treat rev 001 as a prototype: first rides in an open area, helmet on, away from traffic and bystanders.

§11 GENERAL NOTES

- All dimensions mm unless noted. Formulas reference Sheet-0 datums; verify every derived number against the physical pod before cutting.
- Break all sharp edges 0.5 mm. Deburr every bore both sides.
- Fasteners class 10.9 (metric) / Grade 8 (US) minimum at pivot and shock points; all nuts nylock or thread-locked.
- This drawing set describes one pod; mirror-image the carrier plates and repeat for the opposite side of the vehicle. Arms and stack-up are symmetric and do not change.
- Rev 001 supersedes the AI-concept sketch it was derived from. Known deltas: arms changed from single plates to fork weldments (keeps idlers centred on the belt), shock tab moved from 0.45·C to 0.68·C (sane spring rates), tensioner changed from eccentric cam to slot + jack bolt

(easier to fabricate accurately), carrier bearing + anti-rotation link added (the concept had no torque path to the chassis).

- **Rev 001a** (from the 3D fit-check in the companion OpenSCAD model): shocks moved outboard of the carrier plates and the four cross-standoffs replaced by a single keel standoff below the pivot — both original placements collided with the sprocket's swept volume.

APOLLO TRACK POD — ARTICULATED SPLIT-FRAME SUSPENSION · REV 001a · 2026-07-10 · COMPANION 3D MODEL: apollo_track_pod.scad (OpenSCAD) · DRAWN FROM SELLER PHOTOS + PARAMETRIC DATUMS — NOT FROM A MEASURED UNIT. MEASURE, THEN CUT.